

Shining a light on a diagnostic challenge: A case of combined pre- and post-capillary pulmonary hypertension in Shone's Syndrome

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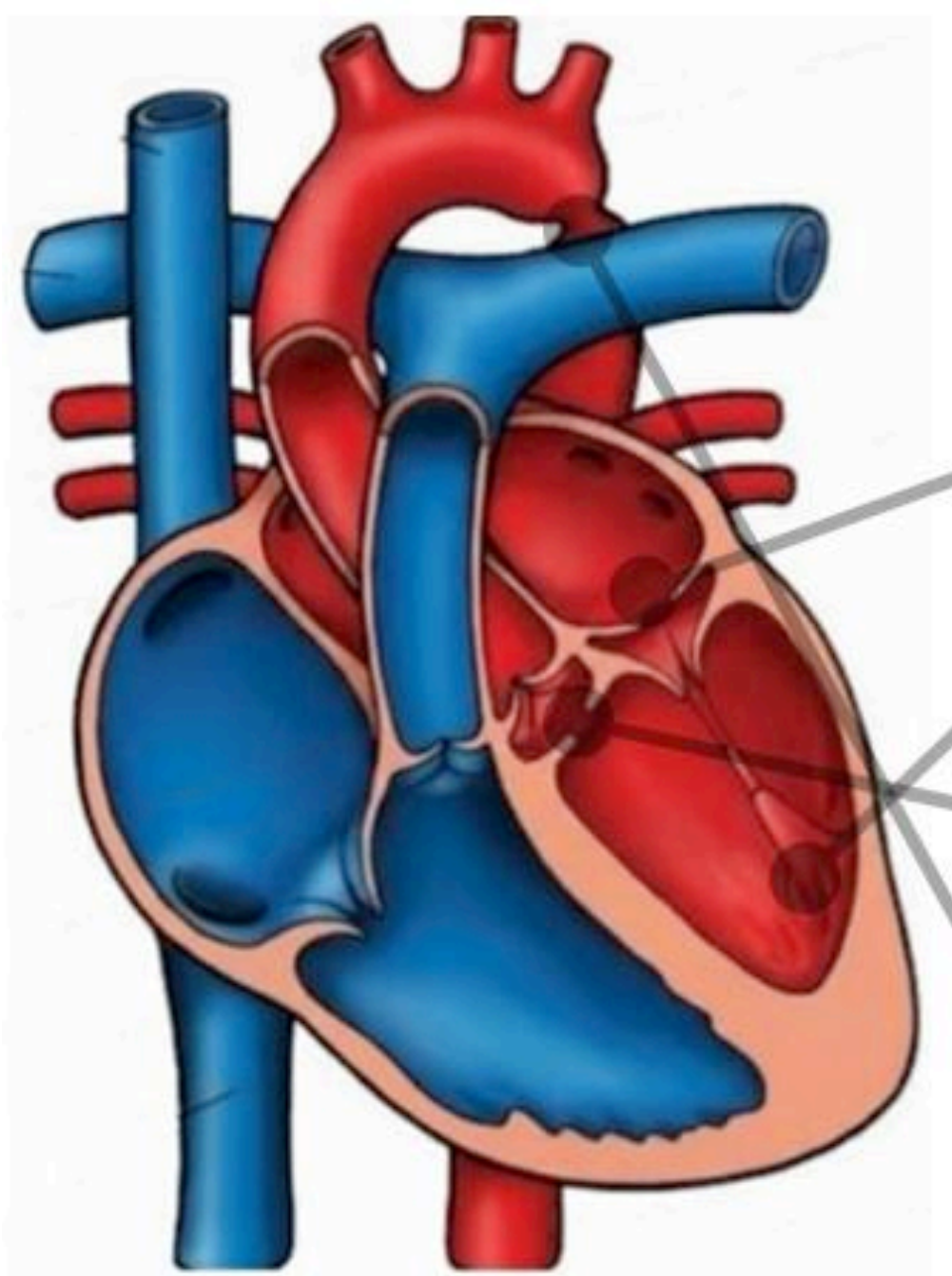


Introduction

- Shone's Syndrome (SS) is a constellation of left-sided congenital lesions.
- Pulmonary hypertension is a serious complication of SS requiring expert multidisciplinary management.
- Combined pre- & post-capillary pulmonary hypertension (cpc-PH) is a complex pathologic phenotype, requiring thoughtful management.

Case Presentation

- 40-year-old female with SS, complicated by severe PH and HFpEF admitted for management of ADHF.
- Past surgical history:**
 - VSD repair + mechanical MVR (1992, Puerto Rico)
 - Redo VSD repair + mechanical AVR (2000, Puerto Rico)
 - CRT-D (2024)
- Prior to admission:**
 - Following with ACHD / PH clinic
 - Serial RHC prior to admission demonstrated severe cpc-PH
 - Tadalafil trial initiated 3 mo prior, supplementing regimen of spironolactone, sacubitril-valsartan, and loop diuretic
- Admission:**
 - RHC: suprasystemic cpc-PH with newly decreased CO prompting upgrade to CICU for PA-C guided IV diuresis and discontinuation of tadalafil
 - TTE and CT demonstrated only modest LA dilation, prompting the clinical question of whether PCWP accurately reflected LAP
 - After achieving euvoemia, repeat RHC with transeptal puncture confirmed elevated LAP; simultaneous TEE demonstrated properly functioning mMVR & mAVR
 - Referred for dual heart / lung transplant evaluation
 - Course complicated by refractory left sided atrial tachycardia; managed with amiodarone, with ablation deferred due to procedural risks a/w mMVR



Components of Extended Shone's Spectrum

1	Cor Triatriatum	Obstructive congenital membrane within LA
2	Supramitral Ring	fibrous membrane above MV, narrowing valve opening
3	Parachute MV	all leaflets attached to single papillary muscle
4	Stiff LV	severe diastolic dysfunction
5	Subaortic Stenosis	membrane or muscular thickening, narrowing LVOT
6	BAV / Small AV Annulus	obstructive AV pathophysiology
7	Small Aortic Arch	underdeveloped without focal narrowing
8	Coarctation of the Aorta	focal narrowing, typically near ductus arteriosus

Figure 1: Components of Shone's Spectrum, per ACHA definition. Original four obstructive lesions bolded with associated anatomic depiction.

	PTA		Admission	
	4 yr prior	7 mo prior	Initial	Repeat
RA	12	11	30	16
RV	67/19	100/14	128/15	89/17
PA	77/49 (65)	100/50 (65)	132/82 (96)	89/36 (55)
PCWP	27	23	35	30 **
Systemic	-	96/55	120/82	111/60
Fick	3.9 (2.2)	4.6 (2.6)	2.6 (1.4)	4.3 (2.5)
TD	4.1 (2.4)	4.2 (2.3)	3.3 (1.8)	-
PVR (Fick)	9.7	9.1	23.5	5.8 **

Figure 2: Serial hemodynamic RHC data. (** Direct LAP measurement)

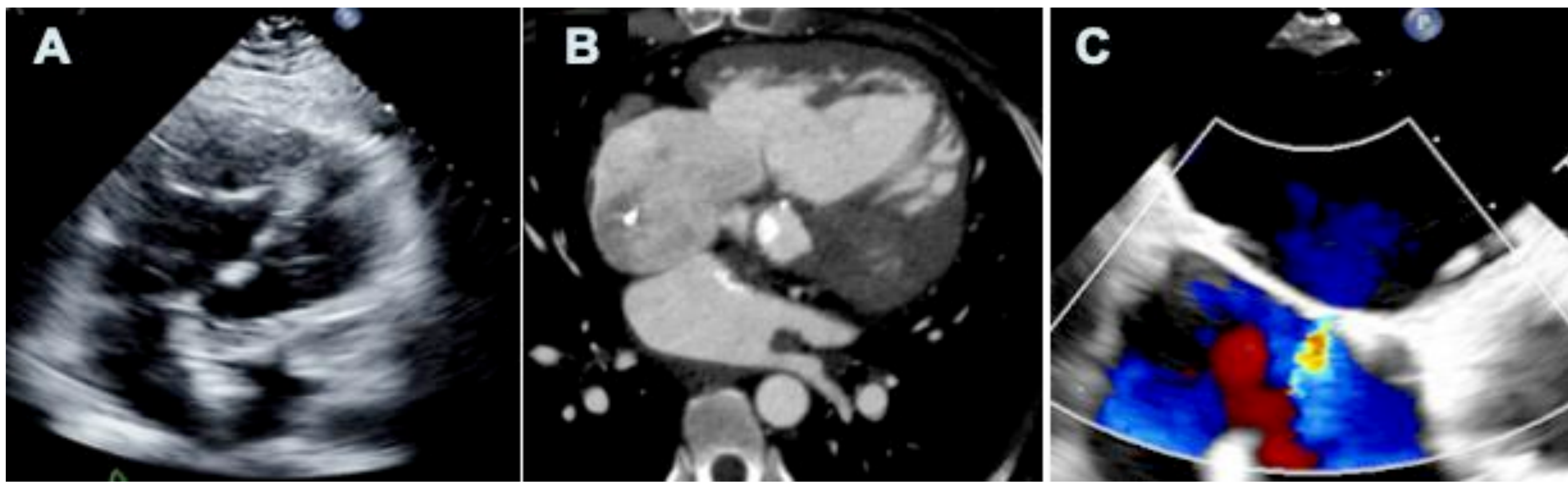


Figure 3: Less than anticipated LA dilation observed on (A) TTE, (B) cardiac CTA, and (C) TEE, with small iatrogenic ASD post transeptal puncture.

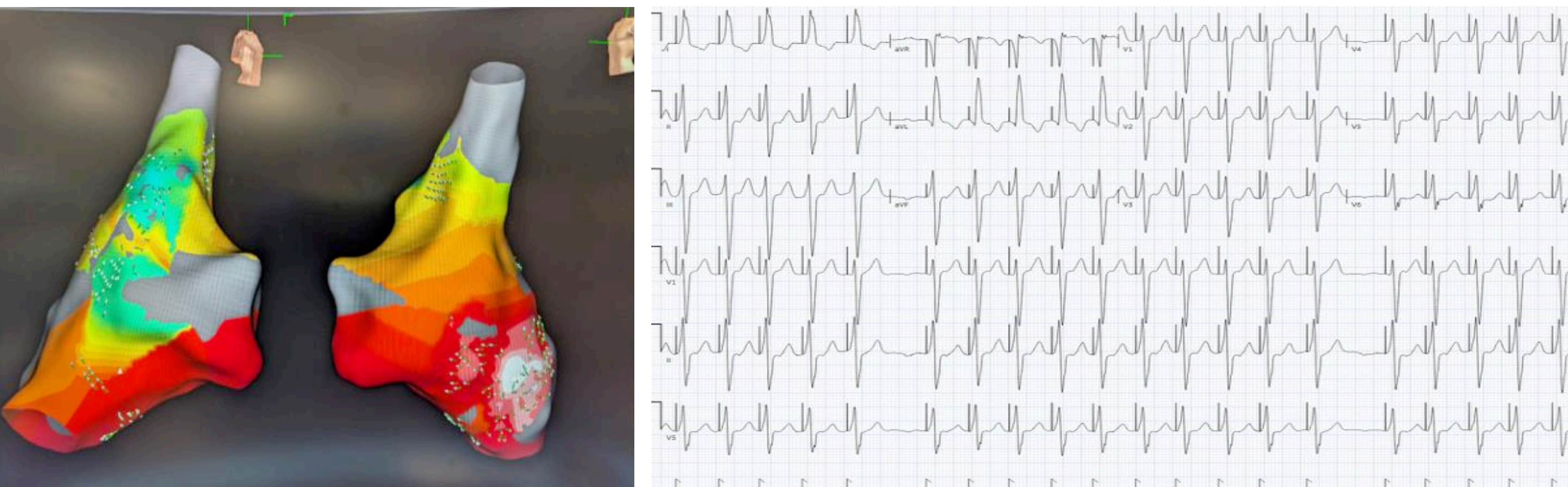


Figure 4: EP study demonstrating left sided AT, with break out near the posterior septum. ECG with paced rhythm and underlying AT.

Discussion

- Shone's Spectrum is a very rare congenital heart disease that presents in infants and children.
 - Prevalence 0.7% of all CHD; unclear etiology, though MV obstruction hypothesized as initial pathologic event during early embryogenesis, resulting in a downstream cascade (demonstrating concept of "no flow, no grow").
 - Relatively low childhood mortality with proper surgical intervention, though patients often require reoperation with a common pathway being eventual transplant.
- This case presents a difficult clinical scenario regarding utility of vasodilator therapy in cpc-PH.
 - Extant trial data holds at best mixed results.
 - Results of the CADENCE trial (sotolol) are highly anticipated within the next year.
- The key consideration in this case revolved around whether her left atrial pressure was truly elevated.
 - If LAP was normal, a trial of alternative pulmonary vasodilator therapy such as IV prostacyclin would have been warranted.
 - With confirmed severely elevated LAP in setting of euvoemia and optimized GDMT for HFpEF, evaluation for heart / lung transplant was most appropriate.

Key Points

- Shone's Spectrum is very rare but has a relatively low childhood mortality with early surgical intervention, though the typical clinical course often results in eventual OHT.
- Vasodilator therapy may play a role in cases of severe cpc-PH but holds a significant risk for decompensation.
- Management of cpc-PH, especially in settings of ACHD, requires specialist case-by-case evaluation.

References

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